BUSA3015 – Business Forecasting, Session 1, 2025

Assessment Task	Professional Practice: Business Forecasting Report
Due date	27th April
Weight (%)	40%
Task description	**From the Unit Guide:** *The purpose of this assessment is to assess students' ability to think critically and solve business problems in the realm of forecasting. In the world of work, you will be expected to communicate your findings via written reports that showcase solutions to business problems while providing a critique of your findings. Students will submit a report which will include a numerical component and a written component. Students will also be asked to submit the accompanying data/calculation file/s. Students can expect to provide an in-person defense of their work during class time.* ***Skills in focus**: Knowledge – Business Forecasting, Data Literacy and Digital Literacy, Critical Thinking, Problem Solving, Written Communication, and Professional Practice.* ***Deliverables**: Numerical answers via iLearn, Report, Data file with working, as well as an in-person defense.* ***This is an individual task.***

Submission Method	**This report will require four separate submissions – one separate submission for each section:** A. (20 Marks). The numerical responses need to be submitted via a quiz tool in iLearn. You will have ten attempts to check whether your Excel output is correct, before you proceed to the written section of the report. Your mark for Section A will be based on your highest mark. B. (60 Marks). The written responses (typed and graphical) need to be submitted via a PDF uploaded via Turn-It-In in iLearn. Your mark for this section will be based on the marking rubric that has been provided. C. (0 Marks). The Excel file that generated the answers and output for **A** and **B** above needs to be submitted. This file will only be looked at for cases which are suspected of violating academic integrity. If you fail to submit your Excel file, your entire assessment (that is all sections) will not be marked. D. (20 Marks). Students can expect to provide an in-person defense of their work during class time. This will be done in classes during the week that immediately follows the submission deadline, that is Week 8 of the session. More details are provided in Section D of this document. If you fail to provide your in-person defense, your entire assessment (that is all sections) will not be marked.
Feedback mechanism(s)	Rubric Individual feedback Cohort-level feedback Sample answers of peers (if a high-performing report's author is willing and able to share)
Feedback available (anticipated date)	2-3 weeks post-submission
Links to Unit Learning Outcomes	On successful completion you will be able to: ▪ Select and apply quantitative and qualitative forecasting techniques for use in business. ▪ Demonstrate practical knowledge of spreadsheets and statistical software to produce business forecasts. ▪ Critically examine business contexts and transform relevant data to provide recommendations for stakeholders.

INTEGRITY MATTERS

Integrity matters – at home, in your workplace and here at University. As a highly valued member of the Macquarie University Community, you carry great responsibility to uphold the good name of our institution. Our reputation is your reputation and will stay with you for life when your degree opens amazing opportunities for you.

If you are ever unsure whether your actions fall within the guidelines of Academic Integrity, please don't hesitate to reach out. Contact the Academic Literacies Unit or your Tutor/Unit Convenor.

ASSESSMENT DESCRIPTION

At the end of this document.

Skills in focus for this assessment

Critical Thinking and Problem Solving
Data Literacy and Digital Literacy
Written Communication

USE OF RESOURCES AND TECHNOLOGIES INCLUDING GENERATIVE ARTIFICIAL INTELLIGENCE

Questions are worded in such a way that you need to critically think of the best way to actually answer the question. Most questions are not direct, and you will need to use your critical thinking skills to figure out the tools / tests / techniques / and more to best answer each question. This is a key employability skill when you enter the workforce. You will be hired on the basis of being able to make these decisions, rather than being micro-managed / micro-directed.

If you use a generative artificial intelligence (AI) tool (such as ChatGPT or similar), without citing the source, you will be penalised for violating academic integrity.

As we have around 400 students in the unit, you also run the risk of plagiarism against other students by using such tools.

If you wish, you may include screenshot/s of any such AI response and then showcase your own response (in typed words) which exhibits your critical thinking where you have modified the AI response to display higher-level thinking skills in line with the unit's learning outcomes.

You are still restricted to the designated page limit.

LATE SUBMISSION

A maximum penalty of five percentage points of the total possible marks will be applied per day to late submissions, for up to a maximum of seven calendar days. Tasks that have not been submitted within the maximum number of additional late days will receive a mark of zero, unless otherwise specified in the late penalties section of the Unit Guide. Late submission for a task will only be permitted when specified in the unit guide. This provision does not apply to online exams or other assessment with a time-limit of less than 24 hours.

Where an application for Special Consideration is approved and the outcome is an extension to the due date of a task, submissions that are received after the new due date will be subject to late penalties that are calculated from the new due date. This only applies where the outcome is an extension to the due date – see the Special Consideration Policy for a schedule of all possible outcomes.

IN-PERSON DEFENSE

This assessment includes an in-person defense in the Week 8 tutorials.

If you submit Section A, B, and/or C by the assessment deadline, but miss your Week 8 tutorial, you will be expected to provide your in-person defense during the Week 8 lecture, including up to two hours after the lecture at the lecture venue if additional time is needed due to number of students who need to provide an in-person defense.

If you submit Section A, B, and/or C by the assessment deadline, but miss **both** the Week 8 tutorial **and** Week 8 lecture opportunity to defend your work in-person for Section D, your entire assessment (that is all sections) will not be marked. If you have a valid reason for missing the opportunities for an in-person defense, you must apply for special consideration.

SPECIAL CONSIDERATION

Students who are granted Special Consideration for the entire assessment will be given a set date for the submission of Sections A, B, and C, as well as a set time and date for Section D. These dates and times are non-negotiable, unless there are circumstances within the allowances of a university policy.

If you submit Section A, B, and/or C by the Special Consideration deadline, but miss the opportunity to defend your work in person for Section D (by the Special Consideration deadline), your entire assessment (that is all sections) will not be marked.

ASSESSMENT INSTRUCTIONS

For the written part of this assignment – the answers must be typed on the pre-formatted DOC file that has been uploaded on iLearn. Your written discussions must not exceed 850 words.

Please do not alter the formatting of the pre-formatted DOC file:

- Do not change the font size.
- Do not change the line spacing.
- Do not change the paragraph settings.
- Do not change the page margins.
- Do not change the headers or footers.
- Do not edit or delete the questions.
- Do not edit any other component of the file apart from typing your answers and cutting and pasting relevant output.

As per the pre-formatted DOC file on iLearn, your answers will be in Calibri font, size 11. The answers are to be in black font.

Not adhering to the above will results in a mark penalty of 10% penalty per page over the limit (or pro rata for partial pages) and/or every 85 words above 850 words (whichever is greater).

A critical thinking skill is about making judgments about the information that is relevant and can be presented in an efficient and effective way.

If you want relevant output to be marked, you can cut-and-paste relevant output into these pages. **Any pages including appendices (beyond the required 4 pages) will not be marked.**

Do not use appendices, all relevant output from Excel or Minitab must be <u>included within the body</u>, within your answer. Appendices will not be marked.

Students have to write their own words to complete the assignment.

All questions about the assignment must be via the iLearn "General discussion forum – Report".

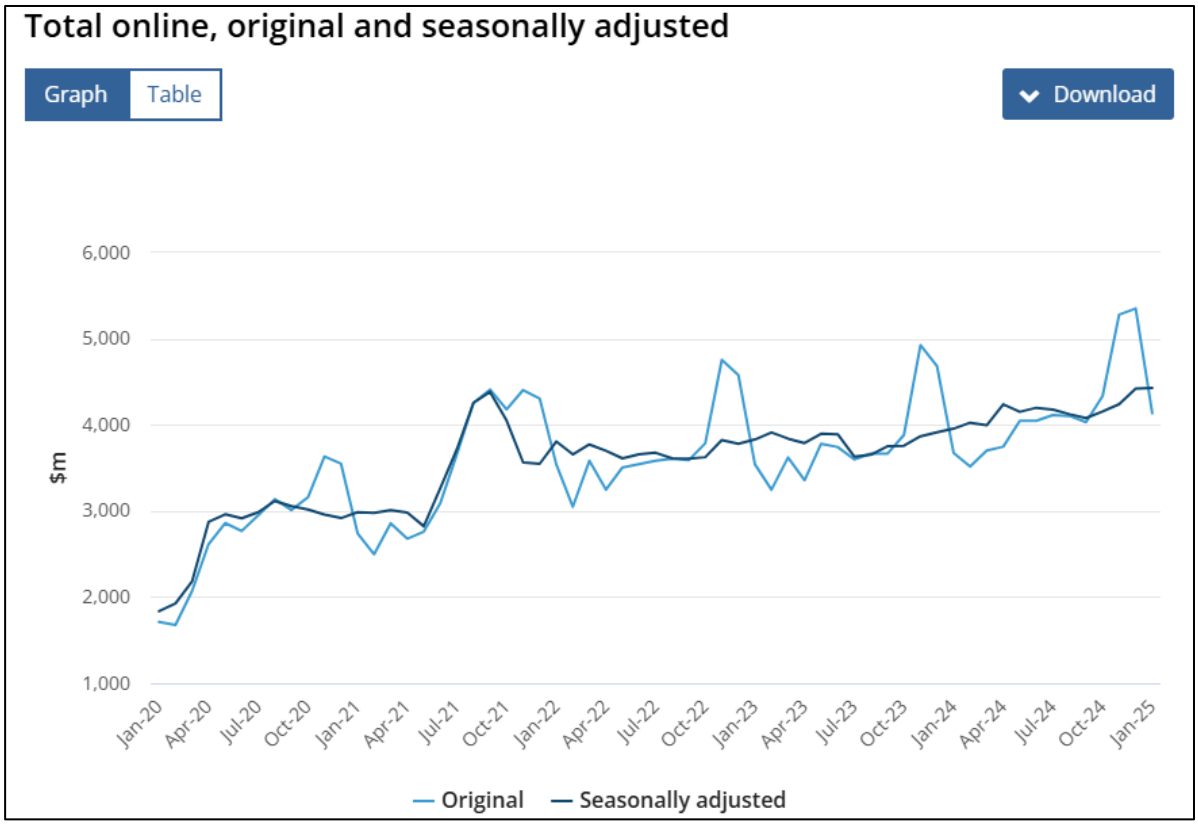
ASSESSMENT DESCRIPTION

You have been employed as a consultant for the *Australian Retailers Association*.

As part of your role in the Business Analytics and Data Analytics team, you have been asked to forecast retail trade for the 12 months from February 2025 to January 2026.

Pre-work to begin this assessment

- Obtain the ABS statistics for *Retail Trade, Australia* - available at:
- https://www.abs.gov.au/statistics/industry/retail-and-wholesale-trade/retail-trade-australia/latest-release#data-download
- Scroll down to *Total online, original and seasonally adjusted.*
- You should see output that looks like this:



- You will see two buttons: 'Graph' and 'Table'. Do not click on 'Download'.
- Click on 'Table'. Copy and Paste the data from February 2022 to January 2025 into Excel.
- Label the columns 'Original' and 'Seasonally adjusted' in your Excel file.
- The ABS website may get updated during the course of the unit, please ignore any data from February 2025 onwards for the purposes of this assessment.

Section A (20 Marks)

Numerical responses to be submitted via a quiz tool on iLearn.

Exercise 1 – Application (10 marks)

Exercise 1 focuses on the ***Seasonally adjusted*** data that you copied above. Forecast the values for February 2025 to January 2026 using Holt's Exponential Smoothing in ***Excel*** with the following parameters: alpha = 0.5 and beta = 0.2. **For the seed of the level use the first observation, Y_1. For the seed of the trend – take the difference of the first two observations ($Y_2 - Y_1$).**

Once you perform Holt's Exponential Smoothing with alpha = 0.5 and beta = 0.2, what are the following numerical values:

1. The within-sample forecast for January 2025.
2. The out-of-sample forecast for July 2025.
3. The out-of-sample forecast for December 2025.
4. The MSE
5. The MAE

Critically think of a way to optimise alpha and beta via the MSE, and report the following values after your optimisation:

6. Alpha
7. Beta
8. The MSE
9. The out-of-sample forecast for July 2025.
10. The out-of-sample forecast for December 2025.

[Please turn over]

Exercise 2 – Application (10 marks)

Exercise 2 focuses on the ***Original*** data that you copied above. Forecast the values for February 2025 to January 2026 using Winters' Exponential Smoothing in ***Excel*** with the following parameters: alpha = 0.5, beta = 0.2, and gamma = 0.1. **For the seeds of the level, trend, and seasonal components – utilise the methods described and discussed in class.**

Once you perform Winters' Exponential Smoothing with alpha = 0.5, beta = 0.2, and gamma = 0.1, what are the following numerical values:

11. The seasonal component for January 2025.

12. The within-sample forecast for January 2025.

13. The out-of-sample forecast for July 2025.

14. The MSE

15. The MAE

Critically think of a way to optimise alpha, beta, and gamma via the MSE, and report the following values after your optimisation:

16. Alpha

17. Gamma

18. The MSE

19. The out-of-sample forecast for July 2025.

20. The out-of-sample forecast for December 2025.

Section B (60 Marks)

Written responses submitted via a PDF upload via Turn-It-In in iLearn:

850 words (+/- 85 words) of written discussion, not counting labels and numbers on graphs AND no more than four A4 sheets in portrait/vertical mode (using the template DOC file provided on iLearn):

Your Section B responses should refer to Exercise 2 from Section A.

In addition to this, you may refer to Exercise 1.

Do not use appendices, all relevant output from Excel or Minitab must be <u>included within the body</u>, within your answer. Appendices will not be marked.

For the model in Exercise 2,

discuss your forecasting method, your forecasts, and the business insights from these, using the following steps:

- Attribution (5 marks)
- Scope (5 marks)
- Application (5 marks)
- Analysis (10 marks)
- Articulation of Issues (10 marks)
- Critique (15 marks)
- Position (10 marks)

You must use the above steps as sub-headings in your response. Failure to do so will result in a loss of marks.

Note in the rubric on iLearn – "sources" are from within the assignment including your own sources of generated results. You do not need to cite the materials provided via iLearn. Given the nature of this task, you will not be penalised for not referring to other sources (although other sources may give you unique insights for your responses). However, in your report, you should consider referring to the information provided by the ABS on the site that is used to download the data

Please turn over for pointers for Section B.

Pointers for Section B

For each of these sub-headings below, at least consider the notes that follow (you can consider more!).

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If you wish, you may include screenshot/s of any such AI response and then showcase your own response (in typed words) which exhibits your critical thinking where you have modified the AI response to display higher-level thinking skills in line with the unit's learning outcomes.

You are restricted to the designated page limit.

You are to use own words throughout the report.

Attribution – Consider the marking rubric.

Scope – Explain the model in Exercise 2 by using language that is understood by a **non-**technical audience. You will need to critically think about whether you discuss the pre-optimised or post-optimised models.

Application – Describe and explain how you applied the data and your knowledge to perform the forecasts in Exercise 2. Describe and explain using language that is understood by a **technical** audience. You will need to critically think about whether you discuss the pre-optimised or post-optimised models.

Analysis – Consider the marking rubric, to assist you, you should include:

A plot of the sample from Exercise 2 and the forecasts (within and out-of-sample) on one chart. You will need to critically think about whether you plot the pre-optimised or post-optimised models. Include description and analysis of the chart, and a description and analysis of your forecast.

Articulation of Issues – Consider the marking rubric, to assist you, you should:

Perform the appropriate check/s and test/s – provide some of this evidence.

What are the issues based on your check/s and test/s above?

Note: we have discussed and conducted several check/s and test/s when we are forecasting in this unit – and it is up to you to determine which checks and tests are appropriate – to determine issues, if any.

Critique – Consider the marking rubric, to assist you, you should:

Critically evaluate your model and critically evaluate the factors you would need to consider when forecasting in light of recent events.

Compare and contrast alternative models.

In the context of business forecasting, critically think and discuss any other considerations that need to be taken into account for your forecasts / forecasting to be useful for business purposes.

Position – Consider the marking rubric, to assist you, you should consider:

This is an informed and justified conclusion that draws upon your discussion above. Given all of your discussion/s above, state your position regarding the business insights to be obtained by your forecasts, by referring to the evidence and ideas that you have discussed above.

Section C (0 Marks)

Submit the Excel file that generated your numerical and graphical responses for Sections A and B using the Excel submission link in iLearn.

While this section is worth zero marks, if you do not submit your Excel file, your entire assessment (that is all sections) will not be marked.

Section D (20 Marks)

In the week of tutorials that immediately follow the submission deadline, that is the Week 8 tutorials, students can expect to provide an in-person defense of their work during class time. Students must perform this in-person defense in their registered tutorial.

If you do not perform your in-person defense, your entire assessment (that is all sections) will not be marked. Please see the In-Person Defense and Special Consideration section of this assignment.

The in-person defense will require responses to questions that ask you to explain how you performed element/s of the assessment while showcasing your file/s. These questions will be basic – to gather information about the integrity of your submission. The questions are not intended to be difficult. This in-person defense is designed to test the academic integrity of your submission. If you have done your own work, you do not need to worry about performing poorly in this section. You do not need to prepare for this in-person defense, other than the fact that you completed Sections A, B, and C in an academically integrous way. **It is envisaged that all students who made a genuine and integrous attempt for Sections A, B, and C, will score full marks for Section D, regardless of their other marks.**